

Anik Parui

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Education

Indian Institute of Technology

Master of Technology in Computer Science and Data Processing; CGPA: 8.51

Kharagpur, India

Aug 2024 - May 2026

Presidency University

Master of Science in Physics; CGPA: 8.4

Kolkata, India

Jul 2020 - Jun 2022

Presidency University

Bachelor of Science in Physics; CGPA: 8.5

Kolkata, India

Jul 2017 - Jun 2020

Skills and Expertise

Programming Languages: C, C++, Standard Template Library (STL), Python, HTML, CSS, JavaScript, SQL

Libraries and Framework: Numpy, Pandas, Matplotlib, Scikit-Learn, Tensorflow, Keras

Tools and Software: VSCode, Jupiter Notebook, GitHub, Google Colab, MySQL

Projects

Multi-model RAG system for Context-Aware Question Answering || [Link](#)

- Developed a full-stack **Multimodal Retrieval-Augmented Generation (RAG)** application that enables users to upload, process, and query multiple document types including PDFs, images, web pages, and videos.
- Implemented an end-to-end retrieval pipeline using **LangChain**, **Hugging Face embedding**, and vector database.
- Implemented semantic search using vector embeddings to retrieve the most relevant document chunks for user queries.
- Integrated **Groq hosted LLM** with custom prompt template to generate precise and context-aware responses.
- Built RESTful APIs with **FastAPI** and developed an interactive frontend using **HTML**, **CSS**, and **JavaScript**.

Grayscale Image Colorization using Convolution Neural Networks (CNNs) || [Link](#)

- Preprocessed the training images by splitting it into **L** and **AB channels**, using L as input and AB as output.
- Reshaped the input data and performed train-test split to separate the training dataset from the testing data.
- Build a CNN architecture with **Convolution**, **Batch Normalization**, **Maxpooling** and **Upsampling** layers.
- Implemented **Adam optimizer** and **MSE loss function** for stable convergence and improved performance.
- Calculated the average **peak-signal-to-noise ratio (PSNR)** to validate model's performance and recorded 48.25 dB.

Predictive Modeling for Credit Risk Assessment using Machine Learning

- Processed missing values, encoded categorical features, and transformed datasets using Label Order Encoding.
- Perform pair plot and **cross-correlation** to identify feature dependencies and visualize relationships via heatmap.
- Applied Log transformation to redistribute data and implemented normalization methods for data preprocessing.
- Performed **Z-score transformation** to detect and remove outliers and applied train-test split to partition the data.
- Applied multiple ML models like **Logistic Regression**, **Random Forest** and **XGBoost** for comparative analysis.
- Analyzed **Confusion matrix** and **Accuracy score** for all models to precisely compare their overall performance.

House Price Prediction Using Linear Regression

- Build a regression model for accurate house price estimation using multi-variable Linear Regression.
- Imputed missing values and applied advanced data visualization techniques like **Distplot**, **KDEplot**, **QQ plot** etc.
- Extracted important features using **PCA** and encoded categorical columns using **OneHotEncoder** technique.
- Performed Standardization for scaling, Power transformation for redistribution, and Z-transformation to detect outliers.
- Evaluated the models' performance by calculating various metrics such as **R2 score**, **Root Mean Square Error**.

Dijkstra Algorithm's Visualizer for Shortest Path using Python

- Build an interactive Dijkstra algorithm visualizer using Python and Streamlit to demonstrate shortest path.
- Developed an interface to add or remove edges and visualize shortest paths with highlighted nodes and edges.
- Developed predefined graphs with customizable options, allowing users to modify them and execute the algorithm.
- Designed responsive UI with custom styling and interactive node and edge visualization with **NetworkX**.
- Optimized Dijkstra's algorithm to accurately compute the shortest path from source node to the target node.

Relevant Coursework

- Data Structure and Algorithm
- Operating Systems
- File Organization and Database Management
- Object Oriented Programming
- Graph Theory and Algorithm
- Artificial Intelligence and Machine Learning

Awards and Achievement

- Secured **AIR 77** in **GATE 2024** in Physics, ranking in the top 0.29% among 26552 candidates.
- Secured **AIR 49** in **CSIR-UGC NET** (Dec 2023) in Physical Science for **JRF**, ranking in the top 0.16% among 30,823 candidates.
- Secured **AIR 354** in **JAM 2020** in Physics, organized by IIT Kanpur.